



Geometric Sequences

Sequences for which consecutive terms have a constant ratio come up in many situations and have interesting properties.

Common Ratios

Definitions

A sequence is a **geometric sequence** if the ratio between consecutive terms is a constant.

$$r = \frac{a_n}{a_{n-1}}$$

The constant r is called the **common ratio** of the sequence.

EXAMPLE 1

Find the common ratio.

A 76, 38, 19, 9.5, 4.75, . . .

SOLUTION

To find the common ratio, select any term of the sequence except the first term. Then divide it by the previous term.

If you select 38, then $r = \frac{38}{76} = \frac{1}{2}$. The common ratio is $\frac{1}{2}$.

► **Think About It** If the common ratio is negative, then the terms of the sequence alternate between positive and negative values.

B $-3, 9, -27, 81, -243, \dots$

SOLUTION

To find the common ratio, select any term of the sequence except the first term. Then divide it by the previous term.

If you select -27 , then $r = -\frac{27}{9} = -3$. The common ratio is -3 . ■

Recursive Rules for Geometric Sequences

Recursive Rule for a Geometric Sequence

The formula for the common ratio $r = \frac{a_n}{a_{n-1}}$ can be rearranged to obtain the following recursive rule for geometric sequences:

$$\begin{array}{ccccc} \text{any term} & \leftarrow & a_n = r \cdot a_{n-1} & \rightarrow & \text{previous} \\ \text{(after first)} & & & & \text{term} \\ & & \downarrow & & \\ & & \text{common ratio} & & \end{array}$$

When you define a geometric sequence recursively, you also need to provide the value for a_1 .

► **Think About It** In the sequence $5, 15, 45, 135, 405, \dots$

$$a_1 = \text{given}$$

$$a_2 = 3a_1$$

$$a_3 = 3a_2$$

$$a_4 = 3a_3$$

and so on.

EXAMPLE 2

For the sequence 8, -16, 32, -64, 128, ... :

A Find a recursive rule.

SOLUTION

Begin by finding the common ratio.

$$r = \frac{g_2}{g_1} = \frac{-16}{8} = -2$$

Substitute -2 for r in the equation $g_n = r \cdot g_{n-1}$.

The recursive rule is $g_n = -2 \cdot g_{n-1}$.

The sequence starts with 8, so $g_1 = 8$.

B Use the recursive rule to find the next two terms in the sequence.

SOLUTION

The next two terms are g_6 and g_7 .

$$\begin{aligned} g_6 &= -2 \cdot g_{6-1} & g_7 &= -2 \cdot g_{7-1} \\ &= -2 \cdot g_5 & &= -2 \cdot g_6 \\ &= -2 \cdot 128 = -256 & &= -2 \cdot (-256) = 512 \end{aligned}$$

The next two terms of the sequence are -256 and 512. ■

► **Think About It** Note that an infinite number of sequences can be defined by the recursive rule $a_n = -2 \cdot a_{n-1}$, such as

-3, 6, -12, 24, ..., and

$$\frac{1}{16}, -\frac{1}{8}, \frac{1}{4}, -\frac{1}{2}, \dots$$

but only one of the sequences with this recursive definition has $a_1 = 8$.

$$8, -16, 32, -64, \dots$$

Writing and Using Iterative Rules for Geometric Sequences

Using the recursive rule, you can see an iterative rule for jumping straight to the n th term in a geometric sequence.

$$a_1 = a_1 \cdot r^0$$

$$a_2 = a_1 \cdot r = a_1 \cdot r^1$$

$$a_3 = a_2 \cdot r = (a_1 \cdot r^1) \cdot r = a_1 \cdot r^2$$

$$a_4 = a_3 \cdot r = (a_1 \cdot r^2) \cdot r = a_1 \cdot r^3$$

... and so on. Notice that each power of r is always 1 less than n .

Iterative Rule for a Geometric Sequence

The iterative rule for the n th term of a geometric sequence with first term a_1 and common ratio r is

$$a_n = a_1 \cdot r^{n-1}.$$

This rule makes it possible to find any term of a geometric sequence, given the first term and the common ratio.

EXAMPLE 3

For the sequence $\frac{1}{64}, \frac{1}{16}, \frac{1}{4}, 1, 4, \dots$:

A Find an iterative rule.

SOLUTION

Find the constant ratio, $r = c_2 \div c_1 = \frac{1}{16} \div \frac{1}{64} = \frac{1}{16} \cdot 64 = 4$.

Substitute $\frac{1}{64}$ for c_1 and 4 for r in the equation, $c_n = c_1 \cdot r^{n-1}$, to obtain the iterative rule.

$$c_n = \frac{1}{64} \cdot 4^{n-1}$$

B Use the iterative rule to find the 10th term in the sequence.

SOLUTION

To find the 10th term, substitute 10 for n .

$$\begin{aligned}c_{10} &= \frac{1}{64} \cdot 4^{10-1} \\&= \frac{1}{64} \cdot 4^9 \\&= 4096 \quad \blacksquare\end{aligned}$$

Graphing Geometric Sequences

EXAMPLE 4

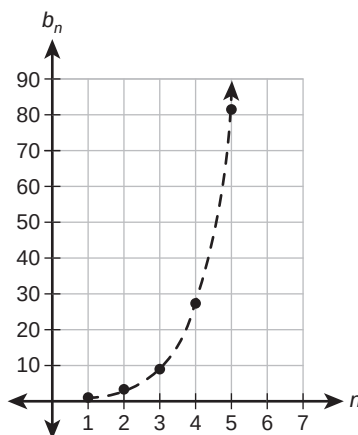
Graph the geometric sequence in the coordinate plane.

A $b_n = 1 \cdot 3^{n-1}$

SOLUTION

The domain is the set of positive integers. Make a table of some points and plot those points to get a partial graph.

n	$b_n = 1 \cdot 3^{n-1}$	b_n
1	$b_1 = 1 \cdot 3^{1-1} = 1 \cdot 3^0 = 1 \cdot 1$	1
2	$b_2 = 1 \cdot 3^{2-1} = 1 \cdot 3^1 = 1 \cdot 3$	3
3	$b_3 = 1 \cdot 3^{3-1} = 1 \cdot 3^2 = 1 \cdot 9$	9
4	$b_4 = 1 \cdot 3^{4-1} = 1 \cdot 3^3 = 1 \cdot 27$	27
5	$b_5 = 1 \cdot 3^{5-1} = 1 \cdot 3^4 = 1 \cdot 81$	81



► **Think About It** The graph of any geometric sequence will consist of discrete points that lie on an exponential curve.

B $k_n = 32 \cdot \left(\frac{1}{2}\right)^{n-1}$

SOLUTION

n	$k_n = 32 \cdot \left(\frac{1}{2}\right)^{n-1}$	k_n
1	$k_1 = 32 \cdot \left(\frac{1}{2}\right)^{1-1} = 32 \cdot \left(\frac{1}{2}\right)^0 = 32 \cdot 1$	32
2	$k_2 = 32 \cdot \left(\frac{1}{2}\right)^{2-1} = 32 \cdot \left(\frac{1}{2}\right)^1 = 32 \cdot \frac{1}{2}$	16
3	$k_3 = 32 \cdot \left(\frac{1}{2}\right)^{3-1} = 32 \cdot \left(\frac{1}{2}\right)^2 = 32 \cdot \frac{1}{4}$	8
4	$k_4 = 32 \cdot \left(\frac{1}{2}\right)^{4-1} = 32 \cdot \left(\frac{1}{2}\right)^3 = 32 \cdot \frac{1}{8}$	4
5	$k_5 = 32 \cdot \left(\frac{1}{2}\right)^{5-1} = 32 \cdot \left(\frac{1}{2}\right)^4 = 32 \cdot \frac{1}{16}$	2

